

Technical Manual: HX-7200 Precision Actuator Controller

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1. Overview

The HX-7200 Precision Actuator Controller is a rack-mounted control unit used to drive high-precision servo actuators in automated manufacturing lines. This manual covers installation, configuration, and troubleshooting for firmware versions 3.0 through 3.2.

2. Installation

2.1 Mounting Requirements

The HX-7200 must be mounted in a standard 19-inch rack enclosure with a minimum of 2U clearance above and below the unit for airflow. Operating temperature range is 0°C to 45°C; ambient temperature above 45°C will trigger the thermal-protection shutdown described in Section 5.3.

2.2 Power Requirements

The unit requires a 24V DC input, rated at a maximum draw of 6.5A under full load. Use only the MDC-approved PSU-2400 power supply; use of third-party supplies voids the warranty described in Section 8.

2.3 Network Configuration

The HX-7200 supports both EtherCAT and Modbus TCP for host communication. Default IP configuration is DHCP; static IP assignment is configured via the front-panel keypad under Menu -> Network -> Static IP.

3. Configuration

3.1 Axis Calibration

Each of the controller's up to 4 axes must be calibrated individually using the auto-tune procedure: Menu -> Axis Config -> Auto-Tune. Auto-tune should be repeated any time the connected actuator or load mass changes by more than 15%.

3.2 Safety Limits

Soft limits (software-enforced travel limits) must be configured for every axis before enabling motion. Hard limits (physical limit switches) are mandatory and cannot be disabled through software, per the safety requirements in Section 6.

3.3 PID Tuning Parameters

Default PID values ($P=1.2$, $I=0.05$, $D=0.01$) are suitable for most standard loads. For high-inertia loads, increase the D term incrementally in steps of 0.01 while monitoring for oscillation.

4. Diagnostic Codes

Code	Meaning	Recommended Action
E-101	Overtemperature shutdown	Verify rack airflow; check ambient temp against Section 2.1
E-204	Encoder signal loss	Check encoder cabling; verify connector seating
E-317	Overcurrent on axis N	Reduce load or verify mechanical binding
E-450	Network communication timeout	Verify EtherCAT/Modbus cabling and host configuration
E-512	Calibration data corrupted	Re-run auto-tune per Section 3.1

5. Maintenance

5.1 Firmware Updates

Firmware updates are applied via USB using the MDC Configuration Utility. Always back up axis calibration data before updating firmware, as updates from version 2.x to 3.x require recalibration.

5.2 Preventive Maintenance Schedule

Inspect connector seating and rack airflow every 6 months. Replace the internal cooling fan every 3 years or upon detection of fan-speed warnings in the diagnostic log.

5.3 Thermal Protection

If internal temperature exceeds 70°C, the unit automatically disables motor output and displays error code E-101 until the unit cools below 60°C.

6. Safety Requirements

Hard limit switches must be installed and wired to the dedicated safety input terminals (SI1–SI4) on every axis. The HX-7200 is rated to Category 3 (per applicable machine safety standards) only when hard limits are correctly wired; without them, the unit must be treated as Category B (no independent safety rating).

7. Ordering Information

Replacement units and accessories reference part numbers HX-7200-BASE (controller), PSU-2400 (power supply), and CBL-ETC-3M (3-meter EtherCAT cable).

8. Warranty

The HX-7200 carries a 2-year limited warranty from date of installation, void if used with non-approved power supplies (see Section 2.2) or if hard limit switches are bypassed (see Section 6).